

You claim that your chargers are like LEGO modules. What's your unique recipe?

We have 30+ intellectual properties in design but this is The One. The innovation that we have got here is in power electronics. You cannot connect two power-generating converters in series, because you would get into load balancing issues. We also faced that problem but we were able to solve it through multiple iterations, and we used a special switching mechanism called GaN FET. We were able to switch GaN FET switches at a frequency of 1MHz, increasing the power density of the charger, which is a great achievement in power electronics systems. With this switching frequency, as well as the hardware we developed, plus the homework we did, we were able to connect the modules in series and parallel to enhance both current and voltage.

Why do you use GaN FET? What's special about it?

We are using GaN FET because we wanted to make sure that we are switching at a frequency of 1MHz or more. If you use MOSFETs and other FETs available in the market, you can't switch at that frequency. If you try to do that, it starts burning. Secondly, when we switch at a higher frequency, the overall size of the power electronics grid will come down. So, we wanted to make a compact and highly efficient power electronics converter and that's why we used GaN FETs.

Was this innovation intentional or accidental?

The initial idea wasn't to use GaN FETs, but the idea came from a need as we had to connect our modules in series and parallel. We were making multiple electric vehicles and designing different battery packs. We realised that we had to

buy different chargers for different sets of batteries. So, we decided to go for one charger for all the batteries, but that type of configuration was not available in the market. We decided to solve this problem ourselves. We brainstormed and arrived at the concept of connecting small modules into series and parallel to arrive at a higher configuration or removing them to achieve a lower configuration. We call this concept an Infinity Charger; there is no limit for voltage and current in this configuration.

What's the secret behind chargers that are half in size?

It is the technology called Immersive Cooling. This tech enables



you to immerse a power-electronics board in a water-like liquid and thereby develop a more compact module. Our charger is half the size of the ones available in the market but can deliver the same capacity of power.

Any safety or structural challenges associated with having power electronics immersed in a liquid? Any risk of short circuits, etc, especially under extreme conditions like an earthquake?

There is no chance of a short circuit because this is a deionised liquid. It is not an electrically conductive liquid. It is only thermally conductive. It is safe. It helps to

protect the battery from thermal runaway. Thermal runaway happens when the battery reaches a particular temperature and catches fire. Using this immersive cooling technology, we don't allow the battery to reach that temperature. The liquid effectively transfers the heat from the switch to the surface. The immersion liquid will get heated up and cool the switch instead. The surface will then get cooled by the surrounding liquid.

Did you face any other challenges in manufacturing such equipment?

Prototyping in India is a bit challenging, of course. Procurement of components is a challenging task. While researching what components to use, we had to try out different types of components available in the market that worked the best for our technology. The other challenge came up during designing. When faced with a problem, we have to go through multiple iterations before we finalise a particular design, and most people don't know when to stop these iterations. They continue to do it because they either want a perfect solution or they think that there is something better. So, the challenge is knowing when to stop and finalise a particular design.

How many members were involved in the development and prototyping of this unique technology?

We had a research team of five members initially who worked for 18 months to come up with this solution. The team has people who have expertise in power electronics and mechanical engineering.

Any mistakes made and/or lessons learnt in the process of developing this tech, which you'd like to share with peer innovators?

Managing the aspirations of the innovators was a challenging task. Research work requires a patient